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Received: 16 October 2018 | Revised: 19 June 2019 | Accepted: 20 June 2019

DOI: 10.1111/desc.12933

PAPER

Developmental Science WILEY

Neural recruitment related to threat perception differs as a function of adolescent sleep

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Funding information
William T. Grant Foundation, Grant/Award Number: 181941; Jeffrey/Wenzel Chair in Behavioral Neuroscience; National Science Foundation Graduate Fellowship

Abstract

Detecting threat cues in the environment is an important aspect of social functioning. This is particularly true for adolescents as social threats become more salient and they navigate increasingly complex relationships outside of the family. Sleep relates to socioemotional processing throughout development, but the neurobiological relevance of sleep for threat perceptions in adolescence remains unknown. In the present study, 46 human adolescents (aged 14–18 years; 26 female) made judgments while undergoing a brain scan about whether unfamiliar, affectively neutral, computer-generated faces were threatening. Prior to the scan, several indices of sleep were assessed nightly for two-weeks using actigraphy. Sleep duration and poor sleep quality (defined as less efficiency, more awakenings, longer awakenings), factors influenced by biological and psychosocial changes during adolescence, elicited distinct neural activation patterns. Sleep duration was positively associated with activation in visual and face processing regions (occipital cortex, occipital fusiform gyrus), and this activation was linked to increased threat detection during the threat perception task. Sleep quality was negatively related to dorsolateral prefrontal cortex activation, which moderated the relation between reaction time (RT) and exposure to faces. Findings suggest reduced threat perception for adolescents with shorter sleep durations and more impulsive responding (as evinced by less consistent RT) for adolescents experiencing worse quality sleep. This study identifies an association between sleep and neural functioning relevant for socioemotional decision making during adolescence, a time when these systems undergo significant development.

KEYWORDS

actigraphy, adolescents, faces, fMRI, sleep, threat

1 | INTRODUCTION

Perceptions of threat are important for guiding social interactions and are often formed by simply viewing someone's face (Bar, Neta, & Linz, 2006). In the absence of sufficient sleep, adults demonstrate reduced discrimination of social cues that convey critical information about social others (Goldstein-Piekarski, Greer, Saletin, & Walker, 2015), decreased ability to attend to stimuli (Lim & Dinges, 2008), and impaired emotion recognition (van der Helm, Gujar, & Walker, 2010), all of which are important for perceiving subtle threat signals in the environment. Biological and psychosocial changes result in normative sleep loss with greater frequency in adolescence compared to other developmental stages (Maslowsky & Ozer, 2014), making adolescents vulnerable to the potentially deleterious effects

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of sleep loss on threat detection. However, the extent to which sleep loss relates to detection of social threats during adolescence, when these types of threats are increasingly salient (Masten et al., 2009), is unknown. In the current study, we tested the hypothesis that adolescents with worse sleep (shorter durations and more disruption) would demonstrate reduced threat perception while evaluating emotionally neutral faces.

Beginning at the onset of puberty, there are significant biological, socioemotional, and neurobiological changes that impair sleep duration and quality (Rofey, McMakin, Shaw, & Dahl, 2013). Sleep durations continue to dramatically decline throughout adolescence (Leger, Beck, Richard, & Godeau, 2012), and reduced sleep has been linked with affective consequences such as negative mood and problematic emotion regulation, as well as health consequences including obesity and inflammation (Baum et al., 2014; Mitchell, Rodriguez, Schmitz, & Audrain-McGovern, 2013; Park et al., 2016; Shochat, Cohen-Zion, & Tzischinsky, 2014). Adolescents also experience disrupted sleep as a result of numerous factors like socializing and stress (Astill, Verhoeven, Vijzelaar, & Someren, 2013; Mazzer, Bauducco, Linton, & Boersma, 2018). Although some work has linked insufficient and problematic sleep with altered neural functioning in adolescents (Beebe, DiFrancesco, Tlustos, McNally, & Holland, 2009; Tashjian, Goldenberg, & Galván, 2017; Tashjian, Goldenberg, Monti, & Galván, 2018; Telzer, Fuligni, Lieberman, & Galván, 2013; Telzer, Goldenberg, Fuligni, Lieberman, & Galván, 2015), there is still much to learn about the mechanisms by which sleep may impact neural systems that support key developmental tasks during adolescence. The limited research on this topic has focused on reward processing (Holm et al., 2009), risk-taking (Telzer et al., 2013), and impulsivity (Tashjian et al., 2017). Although sleep is known to impact neural and socioemotional functioning (Franzen, Siegle, & Buysse, 2008; Talbot, McGlinchey, Kaplan, Dahl, & Harvey, 2010), and developmental changes in sleep and socioemotional systems co-occur during adolescence (Rofey et al., 2013), how sleep-related alterations in neural functioning relate to threat perception has yet to be explored.

The basic ability to evaluate social others requires coordination of neural systems, including a set of "core" and "extended" neural regions (Fairhall & Ishai, 2007; Haxby, Hoffman, & Gobbini, 2000, 2002). These neural systems continue to develop throughout adolescence and into adulthood (Cohen Kadosh, Cohen Kadosh, Dick, & Johnson, 2011; Cohen Kadosh, Johnson, Dick, Cohen Kadosh, & Blakemore, 2013; Golarai, Liberman, Yoon, & Grill-Spector, 2010; Scherf, Behrmann, & Dahl, 2012) and may be vulnerable to stressors like sleep loss (Goldstein-Piekarski et al., 2015). The core network is comprised of brain regions responsible for computing visuoperceptual and cognitive aspects of human faces including the occipital cortex, fusiform gyrus, and superior temporal sulcus (Haxby, Hoffman, & Gobbini, 2000). This core network is present in school-aged children, but activity within core regions has been shown to increase with age with adolescents demonstrating heightened activation when evaluating social counterparts compared to younger children, suggesting neural underpinnings of face processing develop during this time (Aylward et al., 2005; Cohen Kadosh et al., 2011, 2013). The core regions receive inputs from the extended network, which is responsible for social and emotional aspects of face evaluation and is comprised of the amygdala, insula, medial prefrontal cortex, paracingulate cortex, and temporal poles (Fusar-Poli et al., 2009; Gobbini & Haxby, 2007; Said, Dotsch, & Todorov, 2010; Said, Haxby, & Todorov, 2011). These extended regions also undergo substantial development during adolescence (Haist, Adamo, Wazny, Lee, & Stiles, 2013) as socioemotional processing skills are honed (Blakemore, 2008). Prolonged development of these neural systems can be adaptive, particularly during adolescence as complex social inferences such as threat potential become increasingly important (Scherf et al., 2012).

Investigations of the relation between sleep and threat perception in the brain have been limited to research in adults using experimental sleep deprivation approaches (Cohen Kadosh et al., 2011, 2013; Kong, Soon, & Chee, 2011). These studies demonstrate that the psychophysiological systems involved in emotional processes, including threat perception, overlap significantly with those affected by sleep (Dahl, 1996; Walker & van der Helm, 2009). For example, during rapid eye movement (REM) sleep activation in the amygdala, striatum, insula, occipital cortex, and medial prefrontal cortex increases (for a review, Goldstein & Walker, 2014). Of these regions, the amygdala has received the most attention for being robustly implicated in processing of both faces and emotional stimuli (Costafreda, Brammer, David, & Fu, 2008). The amygdala is also activated in conscious and unconscious processing of potentially threatening stimuli (Adolphs, 2008; Öhman, 2005) and this activation is regulated by cortical regions implicated in regulation and control (Kim et al., 2011). These associations have guided prior investigations of threat perception and sleep that largely focus on the amygdala using a priori regions of interest (ROI) analyses. These targeted methods omit potentially important aspects of how adolescent threat perception is neurobiologically affected by sleep.

Despite the lack of research on the links between sleep and threat perception in the developing brain, insights from behavioral literature highlight the importance of sleep on socioemotional

Research Highlights

- Adolescent sleep duration and sleep quality differentially related to neural functioning during a threat perception task.
- Longer sleep durations were associated with greater recruitment of visual and face processing neural regions whereas worse sleep quality was associated with less prefrontal recruitment.
- Greater duration-dependent neural activation related to increased threat detection and blunted quality-dependent neural activation was associated with less consistent reaction times across the task.

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functioning in youth. In a behavioral study of 9–11 years old children, Soffer-Dudek and colleagues (2011) found that greater sleep disruption was associated with poorer performance on an emotion discrimination task (Soffer-Dudek et al., 2011). Research has also shown that adolescents appraise their most threatening worry as significantly more threatening when sleep deprived, compared to when rested (Talbot et al., 2010) and that attentional bias to threat (fearful vs. neutral faces) is associated with poorer sleep in healthy children and young adolescents (Ricketts et al., 2018). This research suggests a link between sleep and threat vigilance in youth, but did not involve adolescents making judgments about whether social stimuli was threatening. This is an important gap in the literature given the changing social environment during adolescence. Adolescents engage in social exploration as they forge complex relationships outside of the family unit (Crone and Dahl, 2012), making other-oriented judgments an increasingly ubiquitous aspect of socialinteraction. Thus, decision-making is a crucial component of socioemotional functioning missing from understanding of sleep and threat perception during development.

In the current study, we tested the association between individual differences in sleep (duration and quality) and neural functioning during afunctional magnetic resonance imaging (fMRI) threat perception task in a sample of human adolescents. During the task, participants made forced choice (yes/no) decisions about whether a series of computer-generated faces were threatening or not threatening (Oosterhof & Todorov, 2008). We assessed several indices of sleep using actigraphy across a two-week period. This approach differs from recent work investigating threat perceptions in sleep-deprived adults (Goldstein-Piekarski et al., 2015) by using naturalistic measures of sleep in adolescents, employing a data-driven whole-brain approach to identify sleep-affected neural systems, and relating neural differences to behavioral judgments. Based on prior work demonstrating distinct effects of sleep duration and sleep quality (defined as sleep efficiency, number of awakenings, duration of awakenings) on neural functioning (Tashjian et al., 2017, 2018) and research demonstrating the relation between sleep and emotion discrimination (Soffer-Dudek et al., 2011; Talbot et al., 2010), we hypothesized these sleep metrics would differentially relate to neural functioning during the discrimination task and that adolescents with longer sleep durations and less disrupted sleep would evince greater acuity in perceiving differences in threat potential among the faces.

2 | MATERIALS AND METHODS

2.1 | Participants

Forty-six human adolescents between the ages of 14 and 18 years are included in analyses (26 female; M_{age} = 15.87 years, *SD* = 1.13 years). Pubertal status information was obtained for 42 participants using the Pubertal Development Scale (PDS; Petersen, Crockett, Richards, & Boxer, 1988). The PDS includes a series of questions about growth spurt, body hair growth and skin change for both boys and girls. Sex-specific questions include body hair growth, facial hair growth, and voice change for males and body hair growth, breast development, and onset of menarche for females. All females who completed the measure (*n* = 25) were postpubertal (100%). Of the males who completed the measure (*n* = 17), one was early pubertal (5.8%), 11 were mid pubertal (64.7%) and 5 were late pubertal (29.4%).

Three additional participants were tested but excluded from analyses. One participant (male, 16 years) did not wear the actigraphy watch, one participant (female, 16 years) completed the study but did not comply with instructions during the task (i.e., did not make button responses), and one participant (female, 17 years) demonstrated excessive head motion of over 4 mm maximum displacement during the scan.

All participants could read and write in English, had normal or corrected vision, and were right-handed. Parent report was obtained to exclude participants with diagnoses of a sleep, developmental or psychiatric disorder, use of psychotropic medication, and medical conditions contraindicated for scanning (i.e., braces). Adolescent participants provided informed written assent and their parent or guardian provided informed written consent. Participants were compensated for their participation. All procedures were approved by the university's Institutional Review Board.

2.2 | Sleep

Sleep indices were tracked using a Micro Motionlogger® Sleep Watch actigraph by Ambulatory Monitoring, Incorporated (AMI). Participants were instructed to wear the actigraph device on their non-dominant wrist at night for 14 consecutive days. Body movement during nighttime sleep was monitored in 1-min epochs using zero crossing mode (ZCM; in ZCM mode frequency of movement is measured by the number of times the signal voltage crosses zero voltage which is then summed over the epoch). Adolescents were asked to push the event marker button when they turned off the lights to go to sleep and again when they got out of bed in the morning. Self-reported sleep and wake times were also collected via daily text messages and daily at-home diary entries to cross-verify actigraphy records. The in-bed period began at the time of the first event marker indicating when participants turned off the lights to go to sleep and ended at the time when the participant awoke in the morning. If event markers were not pressed for a particular night, adolescent report was used as well as other objective measures of sleep collected via actigraph including light and motion readings. Sleep onset time was not recorded until the first of at least three consecutive minutes of sleep and sleep offset time was recorded at last five or more consecutive minutes of sleep. Each nightly record was scored using validated AMI algorithms (Sadeh, Sharkey, &

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Carskadon, 1994, Action4 software package). Actigraphy has been validated for use in adolescent populations (Acebo et al., 1999; Tashjian et al., 2017, 2018) and as a reliable assessment of sleep quality when compared with polysomnography (Marino et al., 2013; Sadeh et al., 1994).

Actigraphy data were used to calculate four sleep indices of interest, *sleep duration*, *sleep efficiency*, *number of awakenings*, and *duration of awakenings*. Sleep duration was calculated by averaging across the 14 days the number of minutes of sleep attained each night, as assessed by actigraphy, from the time adolescents fell asleep to the time they awoke the next morning (average number of days collected per participant M = 12.65 days, SD = 2.65, range 3–14 days). Two participants with less than one week of actigraphy data were excluded from analyses and results remained the same, thus the full sample of 46 is reported here. Consistent with standard use of sleep duration, this measurement included times when participants experienced nighttime awakenings.

Sleep efficiency, number of awakenings, and duration of awakenings were reduced using principal component analysis (PCA). Sleep efficiency was calculated as the percentage of time spent asleep each night (time asleep/sleep duration), with larger percentages reflecting greater sleep efficiency. Number of awakenings experienced each night were averaged across the 14-day duration of the study. For each night, average duration of awakenings was multiplied by the average number of awakenings to calculate a single average of duration of awakenings for each participant. Several well-recognized criteria for using PCA were assessed. First, the items were highly correlated (rs >|.39|, ps < .008). The Kaiser-Meyer-Olkin measure of sampling adequacy was 0.58, above the commonly recommended value of 0.50, and Bartlett's test of sphericity was significant ($\chi^2(3) = 55.22$, $p < .001$). The diagonals of the anti-image correlation matrix were all over 0.5. Given these indicators, PCA was deemed to be suitable. PCA with variance maximizing (varimax) rotation was used to identify and compute composite scores for the components underlying these three sleep metrics. Components were extracted with eigenvalues greater than 1. One component emerged explaining 72.51% of the variance. All items met a minimum criterion of having a primary loading of 0.5 or above and no item had a cross loading of 0.3 or above. Component loadings indicated this component represented poor sleep quality: sleep efficiency (−0.83), number of awakenings (0.78), and duration of awakenings (0.94). Component regression scores were calculated and ranged from −1.32 to 4.16, M = 0.00, SD = 1.00.

Sleep duration and sleep quality have previously been found to differentially relate to neural functioning (Tashjian et al., 2017, 2018). Thus, sleep duration was not included in the PCA and was instead considered separately using raw, demeaned sleep duration values for each participant.

2.3 | Threat perception task

Participants performed a threat perception task inside of the MRI scanner. A set of 70 faces was selected from a published database of computer-generated face stimuli (Oosterhof & Todorov, 2008). Twenty Caucasian face identities were generated using the FaceGen Modeler program version 3.1 (http://facegen.com). Each face identity was then morphed using a threat computer model (Oosterhof & Todorov, 2008) to create seven versions of each identity that ranged in valence from not threatening to increasingly threatening (1–7 scale; Figure 1a). Participants were randomly assigned to a subset of 10 identities. Seven valence versions across 10 identities comprising a set of 70 faces were presented to each participant.

Each of the 70 discrimination trials consisted of a 1.5 s face stimulus presentation, followed by a 2.25 s decision screen instructing participants to press a button in order to classify the preceding face as either "threatening" or "not threatening" (Figure 1b). Trials were pseudorandomly shuffled so that no face identity or threat valence was presented more than twice in a row. Trials began with an interstimulus interval (ISI) fixation cross presented for a jittered duration (maximum duration = 2.3 s, minimum duration = 1.3 s, mean = 1.8 s). After each decision screen a blank screen was presented for 200 ms. Thirty baseline intertrial intervals (ITI) were pseudorandomly interspersed between face trials, during which a fixation was displayed on the screen for a jittered duration (maximum duration = 5 s, minimum duration = 1.5 s, mean 2.5 s). The ISI, blank screens, and ITI served as a baseline condition and provided the contrast for comparison. Inclusion of these events optimized the design for detection of the blood oxygenation level dependent (BOLD) response function (Dale, 1999).

2.4 | Functional magnetic resonance imaging (fMRI) data acquisition

Whole-brain fMRI data were acquired on a 3T Siemens Magnetom Prisma scanner. Parameters for image acquisition were as follows: voxel size = 3.0 mm × 3.0 mm × 4.0 mm, slices = 34, slice thickness = 4.0 mm, repetition time (TR) = 2000 ms, echo time (TE) = 30.0 ms, flip angle = 90 degrees, interleaved slice geometry, field of view (FOV) = 192 mm. AutoAlign was used, which conducts automated positioning and alignment of the anatomy-related slices using anatomical landmarks. Structural images were acquired using a high-resolution, magnetization-prepared rapid-acquisition gradient echo (MPRAGE) sequence for registration purposes (TR: 1900 ms; TE: 2.26 ms; FOV: 250 mm; slice thickness: 1 mm; 176 slices).

2.5 | Functional magnetic resonance imaging (fMRI) analyses

Preprocessing was conducted using FEAT (FMRI Expert Analysis Tool) Version 6.00, part of FSL (FMRIB Software Library, www.fmrib.ox.ac.uk/fsl). Preprocessing consisted of nonbrain removal using BET (Brain Extraction Tool for FSL), high-pass filtering (100-s cutoff), and spatial smoothing using a Gaussian kernel of FWHM 5 mm. The first three volumes were discarded to allow for image stabilization. Motion correction was performed with MCFLIRT

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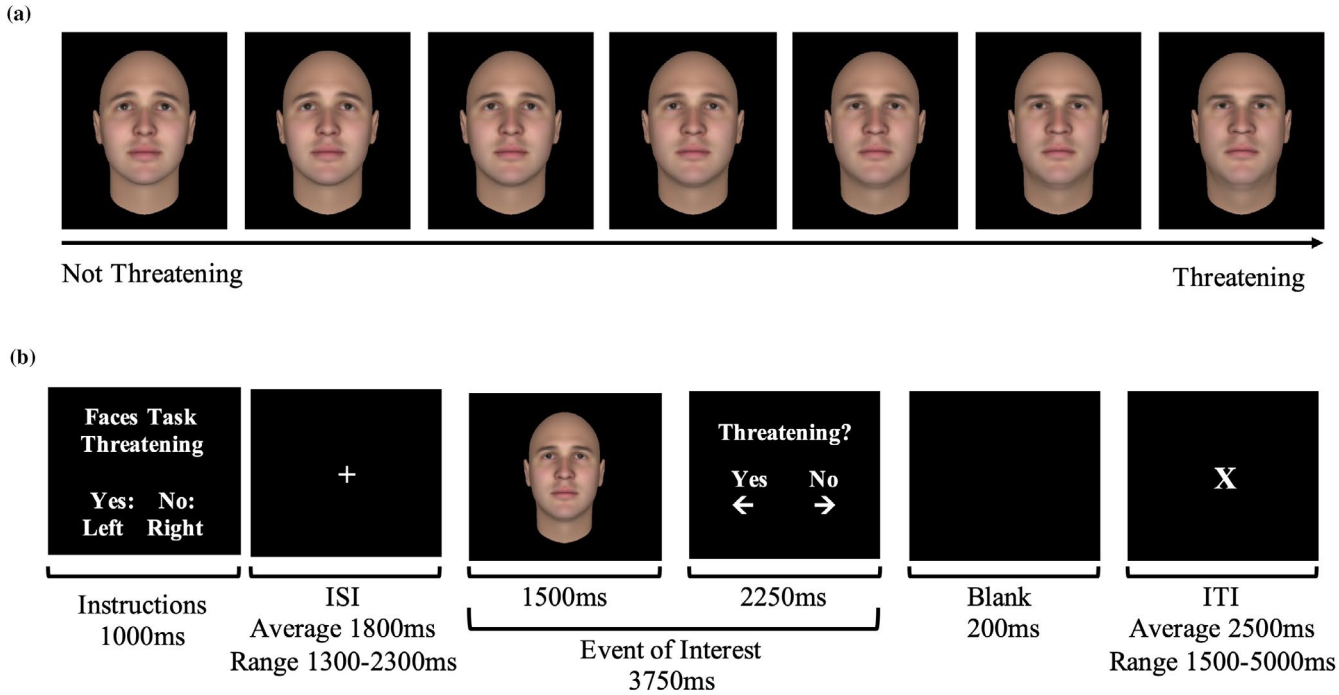

FIGURE 1 (a) Example of the face stimuli gradient for one identity ranging from not threatening to increasingly threatening. (b) Schematic for each trial during the threat perception task. Trials began with an interstimulus interval (ISI) presented for a variable jittered duration (maximum = 2.3 s, minimum = 1.3 s, mean = 1.8 s). After the fixation, a single face stimulus was presented for 1.5 s followed by a 2.25 s response period during which participants were asked to classify the preceding face as either "threatening" or "not threatening" using a button press. Face and decision screens were modeled as the event of interest. After the decision offset a brief blank screen (200 ms) was presented. Pseudorandomly interspersed between trials were intertrial intervals (ITI) presenting a white "X" on a black background for a jittered duration (maximum = 5 s, minimum = 1.5 s, mean = 2.5 s). ISI, blank screens, and ITI events served as the baseline condition contrast to the events of interest and were used to obtain optimal signal estimation (Dale, 1999). Instructions were presented four times throughout the task for three TRs at each presentation and were modeled as events of no interest in face versus baseline analyses. For instructions versus baseline analyses, faces and decision screens were modeled as events of no interest

(intra-modal motion correction tool based on optimization and registration techniques in FSL's registration tool FLIRT) using 24 standard and extended regressors (six motion parameters, the derivatives of those parameters, and the squares of the derivatives and the original parameters). Average absolute displacement was 0.40 mm (range 0.02–2.75 mm), average relative displacement was 0.10 mm (range 0.03–0.67 mm). Analyses were duplicated excluding one participant whose average absolute displacement was over 2 mm (average absolute displacement 2.75 mm, next highest = 1.76 mm) and results remained the same, thus the full sample of 46 is reported here. Each participant's functional data were registered to their MPRAGE using boundary based registration (BBR; Greve & Fischl, 2009) and then to MNI (Montreal Neurological Institute) stereotaxic space with 12 degrees of freedom via FLIRT (FMRIB's Linear Image Registration Tool). Alignment was visually confirmed for all participants. FILM (FMRIB's Improved Linear Model) prewhitening was performed to estimate voxelwise autocorrelation and improve estimation efficiency.

At the individual level, one GLM (Friston et al., 1994) was defined for each participant including regressors for the face events of interest and for instruction screens. The ISI, ITI, and blank screens (Figure 1b) were not explicitly modeled and therefore served as the implicit baseline. Face presentation events were modeled for a duration from face stimulus onset to offset of the decision screen (3.75 s). Events were modeled with a canonical (double-gamma) hemodynamic response function. Temporal derivatives were included as covariates of no interest for all regressors, allowing a better fit for the whole model and reducing unexplained noise.

Group-level whole brain analyses were performed using FMRIB Local Analysis of Mixed Effects (FLAME1) module in FSL (Beckmann, Jenkinson, & Smith, 2003). Sleep duration (demeaned) and sleep quality (standardized coefficients) were entered as regressors in two separate models at the group level (Model 1 and 2, respectively). To determine whether activation differences were specific to face events rather than general physiological effects of sleep (e.g., blood flow, vasoconstriction) we also tested Models 1 and 2 for the contrast of instructions > baseline. An additional model (Model 3) was specified including both sleep duration and sleep quality to assess consistency of effects while controlling for the other sleep variable. Finally, Models 1 and 2 were tested controlling for participant age and sex (demeaned regressors in the whole-brain GLM) (Table S1; Figures S1 and S2). Outliers were de-weighted in the multisubject statistics using mixture modeling (Woolrich, 2008). A cluster-forming threshold of $z > 3.1$ (Eklund, Nichols, & Knutsson, 2016) and an extent threshold of $p < .05$

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familywise error corrected using the theory of Gaussian random fields (Poline, Worsley, Evans, & Friston, 1997) were used. Binarized masks of significant activation were created using *fslmaths* and average contrast parameter estimates were extracted using *fslmeants* for purposes of behavioral analyses. All results are presented on a standard MNI brain and peak coordinates are reported in MNI space. ParaView was used for data visualization (Ahrens, Geveci, & Law, 2005).

2.6 | Behavioral-neural association analyses

To test the relevance of sleep-brain associations for task behavior, effects of neural activation on threat decisions and RT were tested. Behavioral data were analyzed using a multi-level modeling framework because our data consisted of repeated measures (trials) nested within individuals. Behavioral data analyses were performed using R statistical software (version 3.5.0; R Core Team, 2018) and the lme4 (version 1.1-17; Bates, Maechler, Bolker, & Walker, 2015) and reghelper packages. Random intercepts were included in all models.

2.6.1 | Threat decisions

Mixed effects logistic regression was tested using the glmer function in lme4 (link = logit, fitted by Laplace approximation) for all analyses assessing probability of threat decisions. Decisions (1 = threatening, 0 = not threatening) for the *j*th participant (*j*) at the *i*th trial (*i*) were modeled as a function of neural activation extracted from a binarized mask of significant activation from each of the sleep duration and sleep quality fMRI analyses. We estimated the following model for each sleep analysis:

$$\text{Logit}\left(\text{Decision}_{ij}\right) = \gamma_{00} + \gamma_{01}\,\text{Activation}_j + u_{0j} + e_{ij} \quad (1)$$

Post-hoc analyses investigating effects of face valence and neural activation on threat decisions were also tested.

2.6.2 | Reaction time

Mixed effects linear regression was tested using the lmer function in lme4 (lmerTest assessed *t*-tests using Satterthwaite's method) for all analyses assessing RT. Simple slopes analyses were conducted using the simple_slopes function in reghelper. RT (milliseconds) for the *j*th participant (*j*) at the *i*th trial (*i*) was modeled as a function of neural activation extracted from a binarized mask of significant activation from each of the sleep duration and sleep quality fMRI analyses. We estimated the following model for each sleep analysis:

$$\text{RT}_{ij} = \gamma_{00} + \gamma_{01}\,\text{Activation}_j + u_{0j} + e_{ij} \quad (2)$$

Post-hoc analyses investigating effects of stimulus order and neural activation on RT were also tested.

3 | RESULTS

3.1 | Sleep findings

On weekdays, participants went to bed at 11:44 p.m. on average (range 8:15 p.m.–3:42 a.m.) and woke up at 7:21 a.m. on average (range 5:24 a.m.–10:02 a.m.). On weekends, participants went to bed at 12:12 a.m. on average (range 9:09 p.m.–3:38 a.m.) and woke up at 8:27 a.m. on average (range 5:32 a.m.–11:38 a.m.). Participants slept 32.23 min longer on average on weekends than on weekdays.

Actigraphy data were collected for 71.7% (n = 33) of participants during the school year whereas 17.4% (n = 8) were tested over holiday breaks and 10.9% (n = 5) were tested during a combination of school and holiday. Sleep metrics were not significantly related to time of testing (0 = break included, 1 = school only): sleep duration $t(44) = -0.45$, $p = .65$; sleep efficiency $t(44) = -1.87$, $p = .07$; number of awakenings $t(44) = -0.24$, $p = .82$; duration of awakenings $t(44) = 1.37$, $p = .18$.

3.1.1 | Sleep duration

Adolescents attained an average of 435.59 min (7.26 hr) of sleep per night (*SD* = 59.34, range 320.31–563.60), including both weekdays and weekends across the two-week duration of the study. Age, $r(46) = -.25$, $p = .09$, and sex, $r(46) = -.03$, $p = .86$, were not significantly correlated with sleep duration.

3.1.2 | Sleep quality

Average sleep efficiency was 91.16% (*SD* = 5.72, range 77.83%–98.30%), with higher percentages meaning greater time spent asleep compared with time spent awake during the period indicated as nighttime sleep. Adolescents experienced an average of 6.87 (*SD* = 3.94, range 2–25) awakenings per night and those awakenings lasted 26.50 min on average (*SD* = 19.83, range = 4.86–102.00). On weekends compared to weekdays, participants had 0.90 more awakenings that lasted 2.58 min longer.

PCA was conducted to reduce sleep efficiency, number of awakenings, and duration of awakenings into a combined metric of poor sleep quality. Component scores ranged from −1.32 to 4.16, *M* = 0.00, *SD* = 1.00. Age, $r(46) = -.14$, $p = .36$, and sex, $r(46) = -.24$, $p = .11$, were not significantly correlated with sleep quality component scores.

3.2 | Threat perception task: Behavioral results

3.2.1 | Threat decisions

On average, participants rated more faces as not threatening (*M* = 63.38, *SD* = 19.04%) than threatening (*M* = 36.62%, *SD* = 19.04%) with an average discrimination index (% threatening minus % not threatening) of −0.27, *SD* = 0.38. As objective face valence increased

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on the scale from not threatening to threatening, so did the probability of threat decisions, Estimate = 0.40, *SE* = 0.02, *z* = 17.32, *p* < .001. Neither participant age, Estimate = −0.02, *SE* = 0.12, *z* = −0.19, *p* = .85, nor sex, Estimate = 0.45, *SE* = 0.28, *z* = 1.65, *p* = .10, were significantly associated with probability of threat decisions.

3.2.2 | Reaction time

Across the task, participants made their decisions about the faces in 776.52 ms, on average (range 478.59–1024.30). As the task progressed (i.e. as trial order increased), participants demonstrated faster RTs, Estimate = −1.16, *SE* = 0.37, *t* = −3.12, *p* = .001. Older age was associated with faster average RT, Estimate = −49.63, *SE* = 16.01, *t* = −3.10, *p* = .003, but sex was not, Estimate = −22.17, *SE* = 40.23, *t* = −0.55, *p* = .58.

3.3 | Threat perception task: Imaging results

Compared to baseline, face stimuli elicited greater activation in regions previously identified as important for processing social stimuli, including the bilateral amygdala, cingulate gyrus, middle frontal gyrus, bilateral insula, and dorsal striatum. Regions implicated in vision and processing of visual information from the face including the inferior occipital cortex and bilateral fusiform gyrus were also active, as well as numerous other cortical and subcortical structures (Figure 2; Table 1).

3.3.1 | Sleep duration (Model 1)

Demeaned sleep duration values were used as regressors to identify neural activation that varied as a function of sleep duration during the threat perception task. Sleep duration was positively related to activation in the left occipital cortex (including the occipital fusiform gyrus) such that participants with longer sleep durations evinced greater activation magnitude in basic visual and face processing regions while making threat decisions during the task (Figure 3; Table 1; analyses controlling for age and sex are reported in Table S1; Figure S1). Longer sleep durations were also associated with greater activation in the bilateral occipital cortex (including the occipital fusiform gyrus) for the contrast of instructions > baseline (peak activation x = 8, y = −96, z = −2).

3.3.2 | Sleep quality (Model 2)

Component regression scores were used as regressors to identify neural activation that varied as a function of sleep quality. Sleep quality was negatively related to activation in the frontal pole and middle frontal gyrus such that participants with worse sleep quality (less efficiency, more awakenings, longer awakenings) evinced lower activation magnitude in the left dorsolateral prefrontal cortex while making threat decisions during the threat perception task (Figure 4; Table 1; analyses controlling for age and sex are reported in Table S1; Figure S2). Sleep quality was not significantly related to activation for the contrast of instructions > baseline.

3.3.3 | Controlling for sleep duration/sleep quality (Model 3)

Sleep duration and sleep quality component scores were not significantly correlated in this sample, *r* = −.02, *p* = .90. However, an unmeasured variable may account for variance in each of these different metrics of sleep health. Thus, we performed a combined analysis using both sleep duration and sleep quality regressors to examine the stability of the effects of each sleep metric controlling for the other.

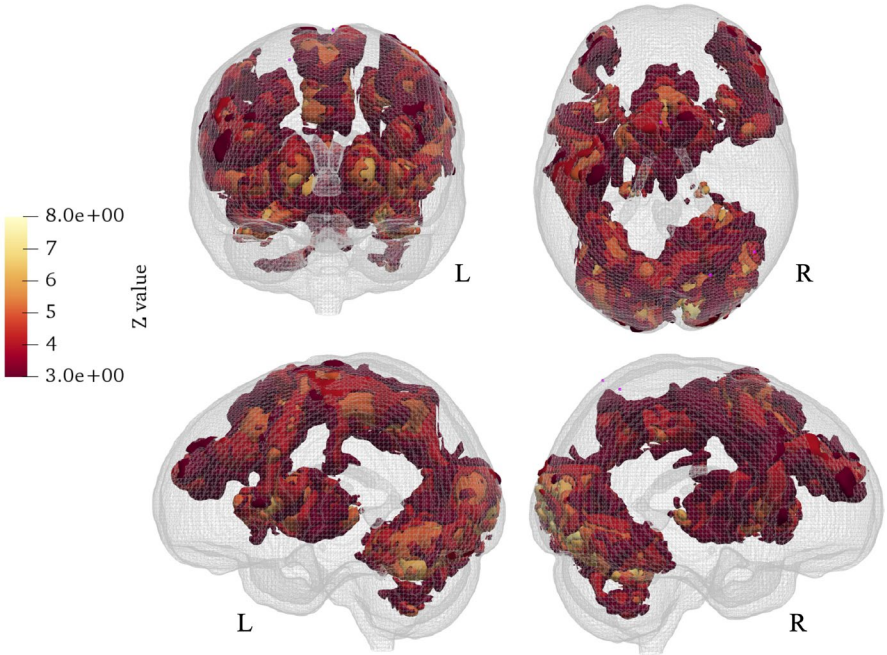

FIGURE 2 Visualization of significant activation for the contrast of faces versus baseline, excluding any sleep variables. Flame1, *z* > 3.1, *p* < .05, outliers deweighted, *N* = 46. Color bar indicates *z* intensity values. L = left hemisphere, R = right hemisphere

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TABLE 1 Significant clusters for the contrast of faces > baseline

Region	R/L	Peak MNI coordinates x	y	z	Max z-value	Voxel count	Volume (mm)
Faces > baseline (no sleep variables)[a]							
Occipital pole	R	18	−96	6	8.54	41,611	332,888
Paracingulate gyrus; Superior frontal gyrus; Juxtapositional lobule cortex	R	2	12	52	7.77	5,058	40,464
Frontal pole	L	−34	52	22	6.32	1,233	9,864
Thalamus	R	24	−28	0	6.74	294	2,352
Cingulate gyrus, posterior	R	0	−24	28	4.30	127	1,016
Sleep duration							
Occipital cortex, inferior; Middle temporal gyrus	L	−48	−68	0	4.66	282	2,256
Occipital cortex, inferior; Occipital cortex, superior	L	−42	−72	4	4.22		
Occipital cortex, inferior; Occipital fusiform gyrus	L	−34	−76	0	4.10		
Occipital cortex, inferior	L	−46	−72	2	4.03		
Occipital fusiform gyrus; Occipital cortex, inferior	L	−30	−78	−2	3.98		
Occipital cortex, inferior; Occipital cortex, superior	L	−36	−76	6	3.92		
Sleep quality							
Frontal pole; Middle frontal gyrus	L	−38	38	32	4.44	107	856
Middle frontal gyrus; Frontal pole; Inferior frontal gyrus	L	−36	36	20	3.87		
Middle frontal gyrus; Frontal pole	L	−32	36	32	3.87		

Note: x, y, and z refer to Montreal Neurological Institute (MNI) coordinates; Voxels refer to each significant cluster; L = left hemisphere and R = right hemisphere. Flame1, $z > 3.1$, $p < .05$, outliers deweighted, $N = 46$.

[a]Local maxima not listed.

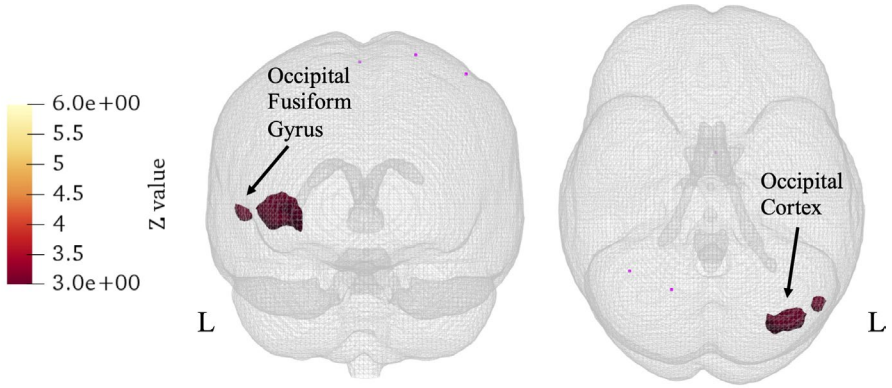

FIGURE 3 Visualization of significantly increased occipital cortex and occipital fusiform gyrus activation associated with longer sleep duration for the contrast of faces versus baseline. Flame1, $z > 3.1$, $p < .05$, outliers deweighted, $N = 46$. Color bar indicates z intensity values. L = left hemisphere

Consistent with findings from the individual analyses, sleep duration was positively related to activation in the left occipital cortex (including the occipital fusiform gyrus) (peak coordinates, x = −48, y = −68, z = 0) and sleep quality was negatively related to activation in the frontal pole and middle frontal gyrus (peak coordinates, x = −38, y = 40, z = 30).

3.4 | Behavioral-neural associations

3.4.1 | Threat decisions

To test whether sleep-related neural activation was associated with participants' threat decisions, we examined linear mixed models described in Equation 1.

Duration-related neural activation to faces was significantly related to decisions (threatening or non-threatening), Estimate = 0.008, $SE = 0.003$, $z = 2.52$, $p = .012$. Participants were more likely to rate faces across the task as threatening if they had greater activation in the occipital cortex (including the occipital fusiform gyrus) (Figure 5). Duration-related neural activation to faces remained significant controlling for neural activation to instructions, Estimate = 0.009, $SE = 0.004$, $z = 2.45$, $p = .014$. Duration-related neural activation to instructions was not significantly related to threat decisions, Estimate = 0.002, $SE = 0.002$, $z = 0.69$, $p = .49$. Quality-related neural activation to faces was not significantly related to threat decisions, Estimate = −0.002, $SE = 0.003$, $z = -0.56$, $p = .57$.

Given the positive association between face valence and threat decisions (also see, Oosterhof & Todorov, 2008), the

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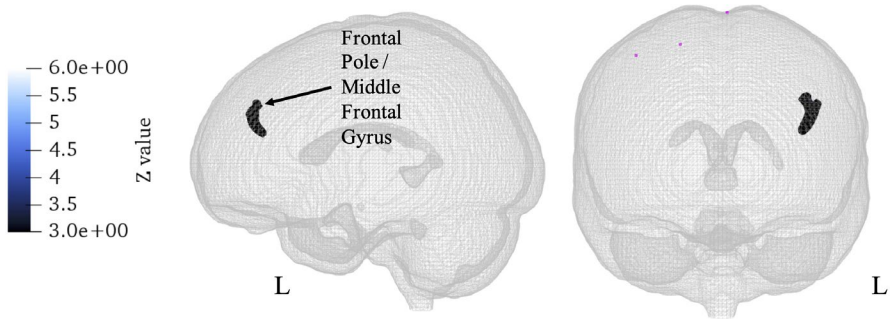

FIGURE 4 Visualization of significantly reduced dorsolateral prefrontal cortex activation associated with worse sleep quality (principal component analysis component) for the contrast of faces versus baseline. Flame1, $z > 3.1$, $p < .05$, outliers deweighted, $N = 46$. Color bar indicates z intensity values. L = left hemisphere

duration-related neural activation effect was tested controlling for objective face valence as follows: $\text{Logit}(\text{Decision}_{ij}) = \gamma_{00} + \gamma_{10}\,\text{Valence}_{ij} + \gamma_{01}\,\text{Activation}_j + u_{0j} + e_{ij}$. As shown in Table 2, duration-related neural activation to faces remained significant controlling for face valence. There was no cross-level interaction between duration-related activation and valence, Estimate = 0.0005, $SE = 0.0006$, $z = 0.80$, $p = .42$, indicating individual differences in participants' threat evaluations were independent of the gradient value of the face.

3.4.2 | Reaction time

To test whether sleep-related neural activation was associated with participants' RT during the task, we examined the linear mixed model described in Equation 2. Neither duration-related activation to faces, Estimate = 0.27, $SE = 0.50$, $t = 0.54$, $p = .60$, nor quality-related activation, Estimate = −0.08, $SE = 0.46$, $t = -0.17$, $p = .87$, was significantly associated with RT.

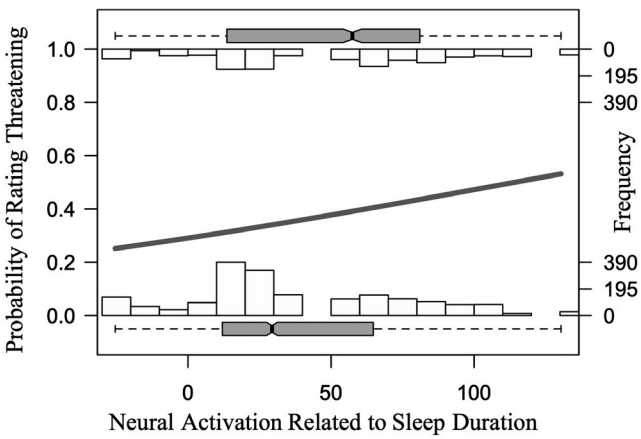

FIGURE 5 Predicted probability plot created in R using logi.hist.plot function in the popbio package (de la Cruz Rot, 2005). Y-axis is probability of threat decisions plotted against neural activation values extracted from the analysis including sleep duration as the regressor of interest (x-axis) (i.e., occipital cortex and occipital fusiform gyrus activation). White boxes are histograms reflecting frequency of binary threat decisions (0 = not threatening, 1 = threatening). Boxplots are displayed in grey and predicted logistic curve is displayed as a trendline. Frequency is the total number of threat decisions for the entire sample

Participants demonstrated faster RTs as the task progressed (i.e. as trial order increased). Interactions between neural activation and trial order were tested as follows: $RT_{ij} = \gamma_{00} + \gamma_{10}\,\text{TrialOrder}_{ij} + \gamma_{01}\,\text{Activation}_j + \gamma_{11}\,\text{Activation}_j{}^{*}\text{TrialOrder}_{ij} + u_{0j} + e_{ij}$. There was a significant interaction with trial order and quality-related activation (Table 3), but not duration-related activation, Estimate = 0.007, $SE = 0.010$, $t = 0.70$, $p = .49$. Simple slopes analyses revealed a significant association between trial order and RT for people with low (−1 *SD*), Estimate = −2.12, $SE = 0.53$, $t = -4.01$, $p < .001$, and average, Estimate = −1.17, $SE = 0.37$, $t = -3.14$, $p = .002$, levels of quality-related neural activation in the frontal cortex, such that they had faster RTs as trial order increased (Figure 6). For adolescents with high (+1 *SD*) levels of quality-related activation, order and RT were not significantly related, Estimate = −0.22, $SE = 0.53$, $t = -0.42$, $p = .67$, indicating consistent RT across the task. Neural activation and sleep quality were inversely associated such that participants with worse sleep quality had lower activation.

4 | DISCUSSION

The goal of this study was to determine whether individual differences in sleep-related neural activation were associated with threat perception in adolescents. For adolescents with shorter sleep durations, recruitment of the occipital cortex (including the occipital fusiform gyrus) was blunted during a task requiring discrimination of

TABLE 2 Fitted linear model of trial-by-trial association between threat decisions and neural activation related to sleep duration, controlling for face valence

Fixed effects	Estimate	*SE*	*z*	Pr(> |z|)
Intercept	−2.717	0.236	−11.497	<0.001
Valence	0.403	0.023	17.318	<0.001
Activation	0.009	0.004	2.466	0.014
Random effects	**Variance**		***SD***	
Intercept	0.912		0.955	

Note: $N = 46$. Observations = 3,010. Threat decisions were coded 0 = not threatening, 1 = threatening. Valence was coded from 1 to 7 increasing in value as threat dimensions increased. Significance was assessed by means of the Wald statistic (z).

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TABLE 3 Fitted linear model of trial-by-trial association between RT and the interaction between order of stimuli presentation and neural activation related to sleep quality

Fixed effects	Estimate	*SE*	*t*	*p*-value
Intercept	840.100	27.310	30.763	<.001
Trial order	−1.707	0.429	−3.975	<.001
Activation	−0.836	0.545	−1.532	.129
Interaction	0.022	0.009	2.541	.011
Random effects		*SD*		
Intercept		125.500		

Note: *N* = 46. Observations = 3,010. Reaction time was coded in milliseconds. Order of stimuli presentation was coded 1 to 70.

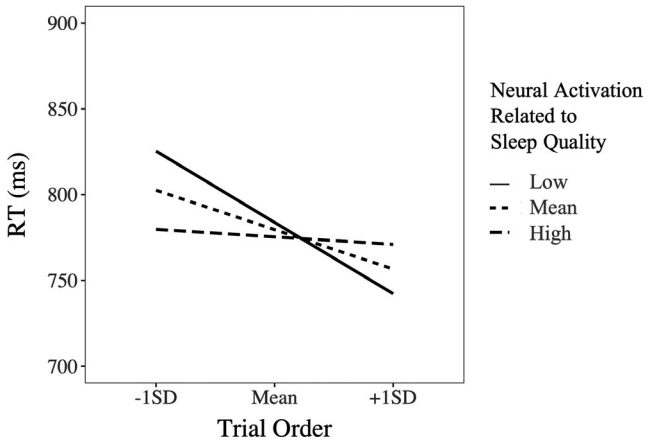

FIGURE 6 Simple slopes analysis showing a significant negative association between reaction time (RT) and trial order (presentation of faces during the task) for those with low (−1 *SD*) and average neural activation in the dorsolateral prefrontal cortex related to sleep quality. For adolescents with high (+1 *SD*) levels of quality-related activation, trial order and RT were not significantly related

threatening from nonthreatening facial stimuli. This blunted activation related to reduced threat detection compared to adolescents with greater neural recruitment in these regions. Poorer sleep quality (defined as less efficiency, more awakenings, longer awakenings) was associated with reduced activation in the dlPFC, a region implicated in attending to threatening stimuli (Sagliano, D'Olimpio, Panico, Gagliardi, & Trojano, 2016), as well as strategic social decision making and impulse control (Steinbeis, Bernhardt, & Singer, 2012; Whelan et al., 2012). This reduced activation interacted with trial order such that adolescents with reduced dlPFC activation to faces demonstrated faster RTs as the task progressed whereas adolescents with greater dlPFC engagement exhibited consistent RTs across the task. These findings highlight the importance of considering multiple facets of problematic sleep and reveal associations between sleep insufficiency and social decision making during adolescence.

Although there were no objectively "correct" responses in our threat perception task, participants differed in their threat evaluations depending on recruitment of visual and face processing neural regions, which was associated with the duration of sleep they attained at night. The interpretation of this differential activation is aided by developmental studies indicating that as individuals age they exhibit increased activation in these regions (Golarai et al., 2010) and greater differentiation among threatening and not threatening stimuli (Lau et al., 2011). In other words, participants showed a more mature profile if they attained longer sleep durations. This is also consistent with literature showing links between cognitive performance and sleep deprivation in adolescents (Louca & Short, 2014) and reduced activation in visual processing neural regions after sleep deprivation (Kong et al., 2011). Adolescents in our sample attained shorter sleep durations than is recommended (National Sleep Foundation, 2014), but our findings suggest that any increase in duration attained may be beneficial for neural and behavioral functioning in sociocognitive domains.

Adolescents with worse sleep quality recruited the dlPFC to a lesser extent during threat discrimination than those with better sleep quality, consistent with prior work on cognitive control and sleep quality in adolescents (Telzer et al., 2013). dlPFC recruitment interacted with trial order such that participants with reduced activation made quicker judgments as the task progressed whereas those with relatively increased activation displayed consistent RTs across the task. Prior work has similarly linked faster RT to threatening stimuli with poor sleep (longer sleep onset delay and increased disruption after sleep onset) in children and young adolescents (Ricketts et al., 2018). Those authors interpreted faster RT to represent anxiety-related bias toward threat. This interpretation aligns with other adolescent research identifying faster RT as a measure of impulsive approach to emotional information (Steinberg et al., 2008). It is worth noting that our RT finding may be task specific. Much of the extant face processing literature involves appraisal of explicitly emotional faces (e.g., Deeley et al., 2008) and links faster RT with greater proficiency in emotion discrimination. In contrast, the present study used emotionally neutral face stimuli with no objective "correct" response. Thus, the ambiguity, and consequent difficulty, of the specific task demands in this study may have required effortful, sustained attention (i.e., no change in RT) to reach a decision threshold. This interpretation is bolstered by findings of attentional deficits observed after disrupted or restricted sleep in youth (Lo, Ong, Leong, Gooley, & Chee, 2016). Our findings extend current understanding of the dlPFC as a neural moderator of sleep and cognitive performance to socioemotional contexts.

This study builds on investigations of face and emotion processing to implicate sleep in functioning of visual processing systems and prefrontal circuitry. An inability to distinguish among facial expressions is associated with severe social dysfunction (Filkowski & Haas, 2017). This is particularly relevant for adolescents as they undergo extensive biological and sociocognitive changes in affective responses to social stimuli (Forbes, Phillips, Silk, Ryan, & Dahl, 2011) while concurrently experiencing chronic insufficient sleep (National Sleep Foundation, 2014; Owens, 2014). Additionally, from childhood to adulthood, the fusiform gyrus shows an increase in volume

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(Golarai et al., 2010) and spatial activation in response to faces (Golarai et al., 2007) and the PFC undergoes substantial functional and structural development (Casey, Jones, & Hare, 2008; Gogtay et al., 2004). Activation in visuoperceptual regions, but not the PFC, were associated with threat perceptions. PFC activation was linked to RT differences across the task, consistent with understanding of the PFC as a crucial component of impulse control. Although, we did not have a sample of adults or children for comparison, the present study identifies links between sleep and functioning of these regions during a period of critical development.

We specifically probed threat as a salient social construct during adolescence. A hallmark of adolescence is increased social exploration (Casey, Getz, & Galván, 2008), which may lead to more exposure to novel faces compared with prior developmental periods. This increased exposure can aid the development of facial expertise (Bukach, Gauthier, & Tarr, 2006), but also requires discernment of potential social threats as adolescents seek out new peer and romantic relationships. Social pursuits in adolescence are inherently risky given that social rejection is one of the most aversive experiences during adolescence (Masten et al., 2009; Sebastian, Viding, Williams, & Blakemore, 2010), and evaluating potential social threats often occurs with limited contextual information (e.g., strangers, novel contexts, neutral expressions). These circumstances make it more difficult and more important to identify subtle differences in facial expressions in order to successfully navigate one's environment. Problematic sleep has previously been linked to deficient emotion regulation and emotion recognition (for a review, Gregory & Sadeh, 2012). For example, after sleep deprivation adults show blunted recognition of affective expressions (van der Helm et al., 2010), which coincides with our finding of reduced threat detection for adolescents with shorter sleep durations. The present study extends understanding of sleep-emotion connections to encompass adolescent socioemotional decision making and indicates naturalistic differences in sleep are related to neural functioning when evaluating emotionally neutral faces.

Prior work in sleep deprived adults using the same face stimuli in this study has linked sleep deprivation with increased subjective threat ratings (Goldstein-Piekarski et al., 2015) whereas the current study found longer sleep durations were associated with increased threat ratings. Several reasons for this disparity are possible: our sample is approximately 2.6× larger, our study focused on adolescents who as a group made more evaluations of threat (36.62%) than the adult sample (24.0% in sleep-rested condition, 29.2% in sleep-deprived condition), and we used naturalistic measures of sleep duration and sleep quality to investigate inter-individual differences in sleep-associated neural response to faces whereas Goldstein-Piekarski and colleagues (2015) were testing intra-individual differences after experimentally imposed sleep deprivation. The current study also utilized a whole-brain analytical approach to identify sleep-related neural functioning rather than extracting activation using ROIs as does much of the prior literature investigating threat perception. Future work with a larger age range spanning childhood to adulthood would be informative for understanding age and sleep related differences in threat perception.

The current findings should be considered in the context of certain limitations. Although high school has been identified as a period of increasingly problematic sleep making it an ideal age to study sleep-brain associations, we cannot determine whether these findings generalize to other age groups. The stimuli in this study were computer-generated male, Caucasian faces. We did not have sufficient power to investigate effects of sex- and race-related factors known to be relevant for face evaluation. As such, future work may seek to explore how sleep interacts with other factors germane to trait inferences. Naturalistic differences in sleep were assessed using actigraphy over a two-week period. Acute sleep restriction may result in different effects for neural functioning and threat evaluation. Although interpretation of the current findings is supported by nationally recommended sleep standards, the present study is not optimized to make determinations as to whether participants in this sample achieved insufficient sleep. Sleep quality was estimated using PCA dimension reduction of three commonly used metrics of sleep quality to provide a more complete assessment of duration- and awakening-related contributions to sleep quality. Considerations of sleep quality are relatively burgeoning in developmental neuroscience compared to investigations of sleep duration. Future work with experimental manipulations of sleep may consider probing potential differential weighting of these factors to quality-related differences in neural functioning.

The ability to detect potential threats is an important social task during adolescence. Perceptions of threat can inform approach and avoidance behaviors, making it important to understand individual differences that contribute to variability in threat perception. Sleep-related differences in neural activation were associated with threat detection and RT during a threat perception task. Differential contributions of sleep duration and sleep quality illustrate the complexity of sleep and the importance of evaluating naturalistic differences during a developmental period of dramatic change. Our findings suggest development of expertise in face processing and socioemotional decision making may be affected by normative sleep deficiencies during adolescence and that these sleep-related problems relate to perceptions of threat. Future research is needed to understand how these associations extend to other social decision-making contexts.

ACKNOWLEDGEMENTS

This work was supported by a Scholars Award from the William T. Grant Foundation and the Jeffrey/Wenzel Chair in Behavioral Neuroscience to Dr. Adriana Galván. Preparation of this manuscript was supported in part by the National Science Foundation Graduate Fellowship to Sarah M. Tashjian.

CONFLICT OF INTEREST

The authors declare no conflict of interests.

DATA AVAILABILITY STATEMENT

Data and materials are publicly available on the Open Science Framework (OSF; https://osf.io/7um6q/).

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How to cite this article: Tashjian SM, Galván A. Neural recruitment related to threat perception differs as a function of adolescent sleep. *Dev Sci*. 2020;23:e12933. https://doi.org/10.1111/desc.12933

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